

Lab #:

Date:

Title: Gravimetric determination of water in hydrated Barium Chloride

Topic: Gravimetric Analysis

Aim: To determine the degree of hydration of barium chloride using gravimetric analysis.

Appartus/Materials:

Diagram:

Method:

1. Place a crucible and lid on a clay triangle and heat over the Bunsen Burner for 10 minutes.
2. Use a pair of Tongs to remove the crucible and lid and place in a desiccator and allow to cool to room temperature.
3. Use a top-loading balance to measure the mass of the empty crucible and lid and record the results in a table.
4. Add approximately 2 g of the hydrated salt into the crucible and heat gently for the first 2 minutes then strongly for another 10 minutes over the Bunsen Burner.
5. Remove the crucible and its contents from the flame and allow to cool in the desiccator until it reaches room temperature.
6. Reweigh the crucible and its contents on the top-loading balance and record the results in the table.
7. Repeat the heating and cooling cycles until there is no further change in mass of the crucible and its contents.

Results:***Mass of hydrated barium chloride***

Mass of empty crucible + lid (g)	
Mass of crucible + lid + hydrated salt (g)	
Mass of hydrated salt (g)	

Trial	1	2
Mass of crucible + lid + anhydrous salt		

Calculations:

1. Calculate the mass of the anhydrous salt
2. Calculate the moles of the anhydrous salt
3. Calculate the mass of water in the hydrated salt
4. Calculate the moles of water lost
5. Determine the mole ratio between the anhydrous salt and water, hence, the degree of hydration.

Discussion:

- What physical changes were observed as the barium chloride hydrate was heated?
- How did the mass of the sample change after heating?
- Why was the crucible reheated and weighed until a constant mass was obtained?
- What is meant by the “degree of hydration” of a salt?
- Based on the calculations, what is the most likely formula of the hydrated salt?

- Does the experimental value for the degree of hydration match the known value?

Explain any discrepancy.

- How might incomplete removal of water, or overheating, affect the calculated degree of hydration?

Sources of Error/Limitations:

Precautions:

Assumptions:

Conclusion: